
Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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Team ID: SWTID-2026-2214

PHASE 6 – PROJECT TESTING

1. Introduction

The Project Testing phase verifies whether the implemented ServiceNow ACL system works according to the requirements.

Different users, roles, operations, and branch records were tested to ensure that authorized users can access permitted records while unauthorized users are restricted.

2. Testing Objectives

The main objectives of testing are:

- To verify the Read ACL.
 - To verify branch-based record restriction.
 - To verify the Create ACL.
 - To verify the Write ACL.
 - To verify the Delete ACL.
 - To verify administrator access.
 - To verify unauthorized user restrictions.
 - To ensure the ACL script works correctly.
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3. Test Environment

The testing was performed using:

- ServiceNow instance
 - Institution Details table
 - EEE, ECE, and CSE sample records
 - Administrator account
 - EEE test user
 - Custom roles bb1, bb2, bb3, and bb4
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4. Test Cases

Test ID	Test Scenario	Expected Result	Result
TC01	Admin accesses EEE record	Access allowed	Pass
TC02	Admin accesses ECE record	Access allowed	Pass
TC03	Admin accesses CSE record	Access allowed	Pass
TC04	bb1 user accesses EEE record	Access allowed	Pass
TC05	bb1 user accesses ECE record	Access restricted	Pass
TC06	bb1 user accesses CSE record	Access restricted	Pass
TC07	User without required role attempts access	Access denied	Pass
TC08	bb2 user creates a record	Create allowed	Pass
TC09	User without bb2 attempts to create	Create restricted	Pass
TC10	bb3 user edits a permitted record	Edit allowed	Pass

TC11	User without bb3 attempts to edit	Edit restricted	Pass
TC12	bb4 user deletes a permitted record	Delete allowed	Pass
TC13	User without bb4 attempts to delete	Delete restricted	Pass

5. Read Access Testing

Test 1 – Administrator

User: Administrator

The administrator attempted to access records from all three branches.

- EEE: Allowed
- ECE: Allowed
- CSE: Allowed

Expected Result: Administrator should have full access.

Result: Pass

Test 2 – EEE User with bb1

The EEE user with the bb1 role attempted to access records.

- EEE: Allowed
- ECE: Restricted
- CSE: Restricted

Expected Result: The user should be able to access EEE records according to the Read ACL and branch condition.

Result: Pass

Test 3 – Unauthorized User

A user without the required Read role attempted to access the Institution Details records.

Expected Result: Access should be denied.

Result: Pass

6. Create Access Testing

The Create ACL requires the bb2 role.

Test

A user with bb2 attempted to create an Institution Details record.

Expected Result: Record creation should be allowed.

Result: Pass

A user without bb2 attempted the same operation.

Expected Result: Record creation should be restricted.

Result: Pass

7. Write Access Testing

The Write ACL requires the bb3 role.

Test

A user with bb3 attempted to modify an Institution Details record.

Expected Result: Editing should be allowed according to the configured access rules.

Result: Pass

A user without bb3 attempted to modify a record.

Expected Result: Editing should be restricted.

Result: Pass

8. Delete Access Testing

The Delete ACL requires the bb4 role.

Test

A user with bb4 attempted to delete an Institution Details record.

Expected Result: Deletion should be allowed according to the configured access rules.

Result: Pass

A user without bb4 attempted to delete a record.

Expected Result: Deletion should be restricted.

Result: Pass

9. Security Testing

The following security checks were performed:

- Unauthorized users were prevented from accessing restricted records.
- EEE users were restricted from non-EEE records under the Read condition.
- Create permission was controlled using bb2.
- Write permission was controlled using bb3.

- Delete permission was controlled using bb4.
 - Administrators retained full access.
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10. Test Summary

Area Tested	Status
User Authentication	Pass
Role Verification	Pass
Read ACL	Pass
Branch Restriction	Pass
Create ACL	Pass
Write ACL	Pass
Delete ACL	Pass
Administrator Access	Pass
Unauthorized Access Restriction	Pass

11. Testing Outcome

The testing confirmed that the configured ACLs successfully control access to the Institution Details table.

The system provides different permissions based on the user's assigned role, while the Read ACL uses the Branch = EEE data condition for the intended branch-level restriction.

12. Conclusion

The Project Testing phase verified the functionality and security of the Script-Controlled ACL system. The test cases demonstrated that authorized users could perform their permitted operations, while unauthorized users are restricted.
